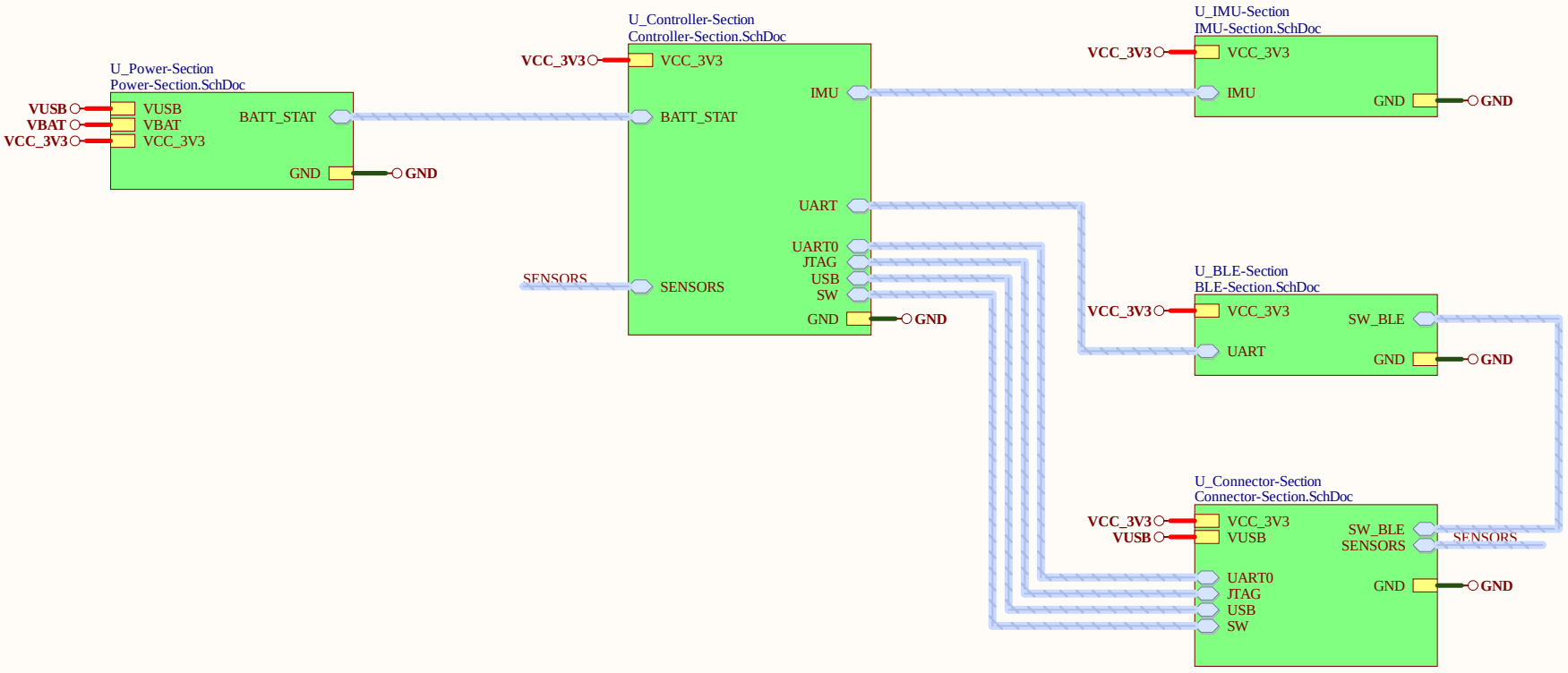
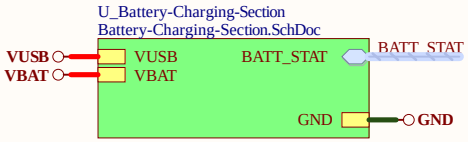
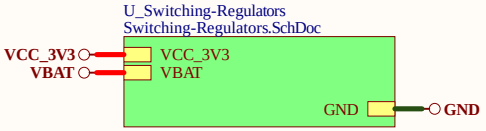
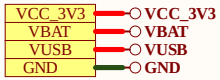


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: ArmBand-Control-Board.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 2-10-2025	Sheet 1 of 8	
	File: D:\eSpark-Consultants\..\Top-Sheet.SchDoc	Drawn By: Huzaifa Ali	









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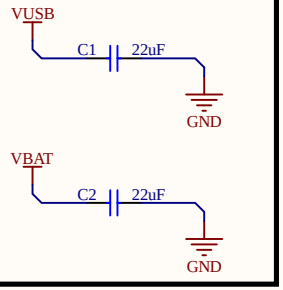
Title	ArmBand-Control-Board.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	2-10-2025	Sheet 2 of	8
File:	D:\eSpark-Consultants\..\Power-Section.SchDoc	Drawn By:	Huzaifa Ali

BATTERY STATUS

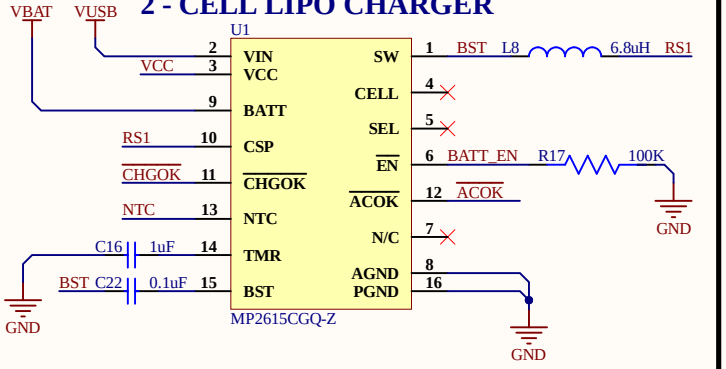


CELL is floating for 2 cells
SEL pin is Floating for 4.2/cell terminating voltage
IC=52.5mV/Rs1 mOhm
IC=2.1A
NTC sensing not used.

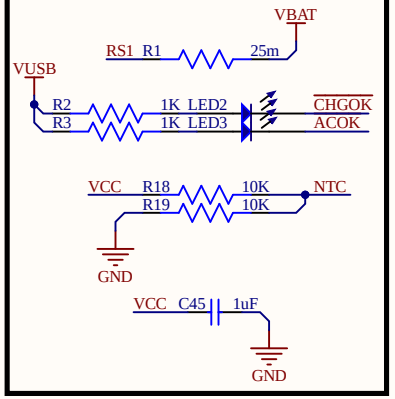
DECOUPLING CAPS

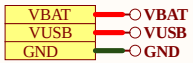


2 - CELL LIPO CHARGER



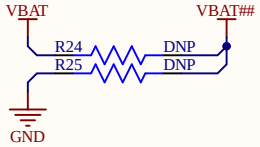
SUPPORTING CIRCUIT



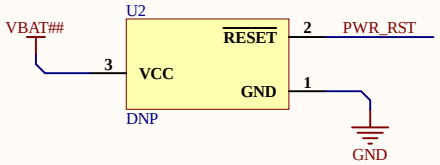


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	Size: A4	Number: 1	Revision: 1
	Date: 2-10-2025	Sheet 3 of 8	
	File: D:\eSpark-Consultants\..\Battery-Charging-Section.SchDoc	Drawn By: Dr. Zaifa Ali	

VDR FOR 6V to 2.9V

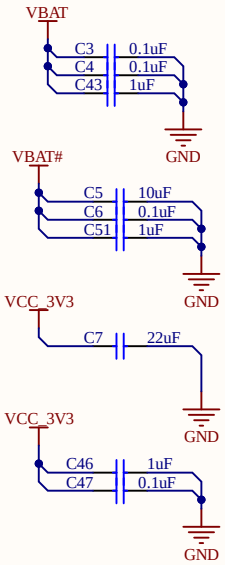


2.9V BATTERY SUPERVISOR

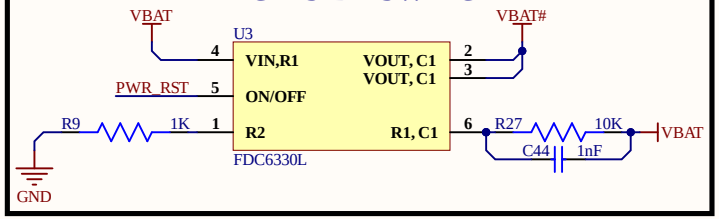


RESET Output remains low while VCC is below the reset threshold, and for at least 140ms after VCC rises above the reset threshold.
Datasheet pg 5

DECOUPLING CAPS

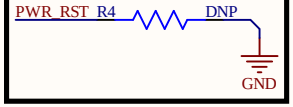


HIGH-SIDE SWITCH

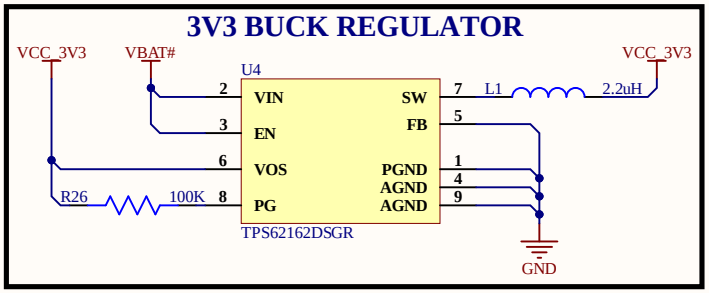


U3 will turn off when ON/OFF is low.
U3 will be enabled when ON/OFF is high.
VON/OFF= 1.5V-8V
VIN= 3-20V

R-PULL DOWN

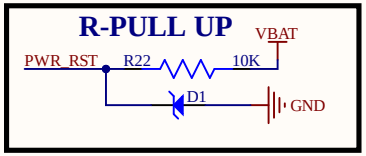


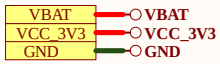
3V3 BUCK REGULATOR



Fixed regulator.
VOUT= 3V3/1A

R-PULL UP





Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	ArmBand-Control-Board.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	2-10-2025	Sheet 4 of	8
File:	D:\eSpark-Consultants\..\Switching-Regulators.SchDoc	Drawn By:	Huzaifa Ali

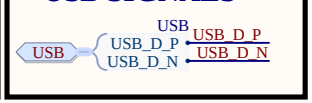
BATTERY STATUS



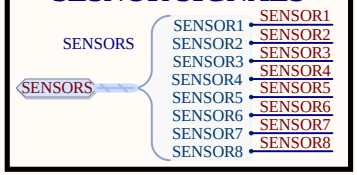
IMU SIGNALS



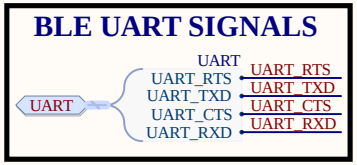
USB SIGNALS



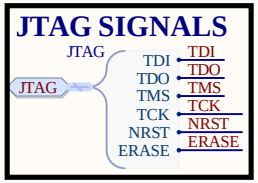
SESNOR SIGNALS



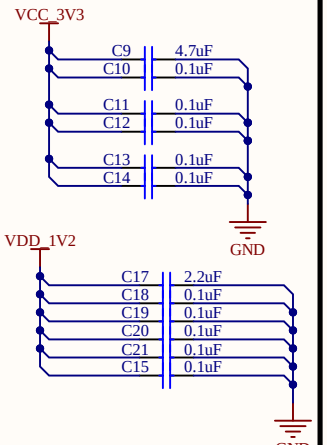
BLE UART SIGNALS



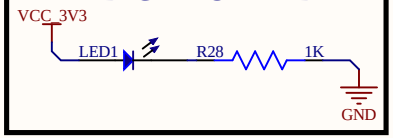
JTAG SIGNALS



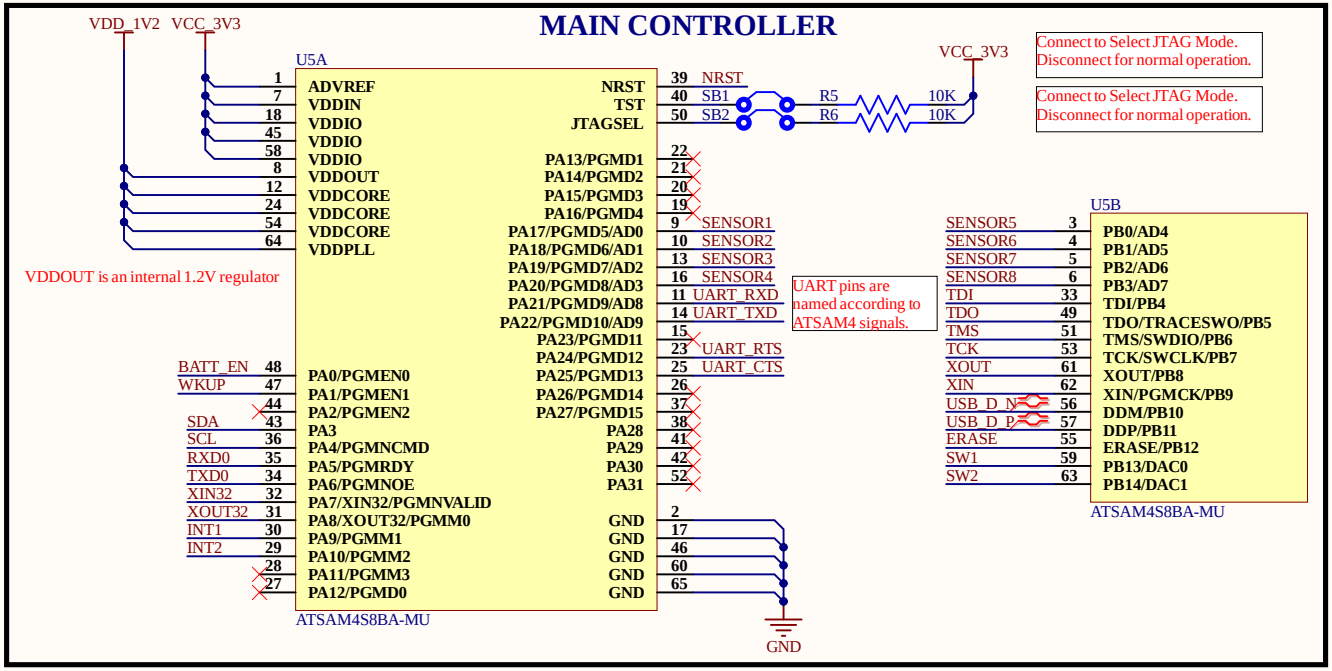
DECOUPLING CAPS



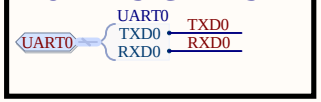
INDICATION LED

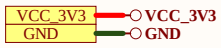


MAIN CONTROLLER

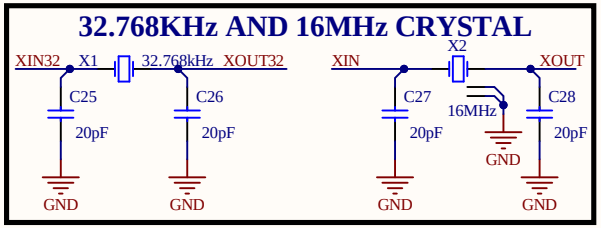


UART SIGNALS

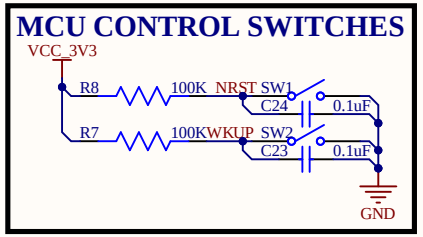




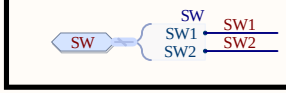
32.768KHz AND 16MHz CRYSTAL



MCU CONTROL SWITCHES



SWITCH SIGNALS



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Title	ArmBand-Control-Board.PrjPcb	
Size: A4	Number: 1	Revision: 1
Date: 2-10-2025	Sheet 5 of 8	
File: D:\eSpark-Consultants\..\Controller-Section.SchDoc	Drawn By: Huzaifa Ali	

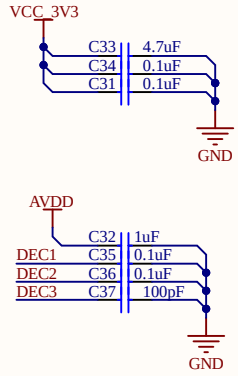
BLE UART SIGNALS



SW SIGNALS

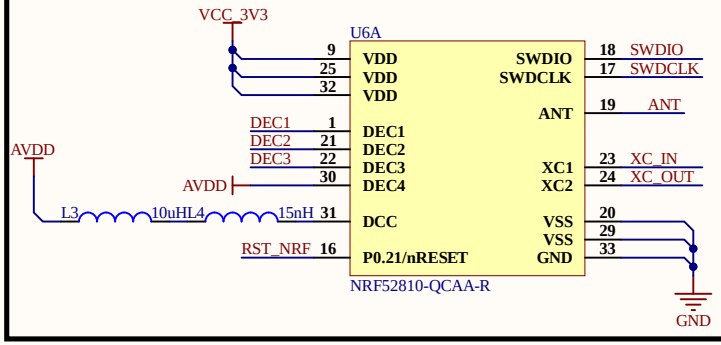


DECOUPLING CAP

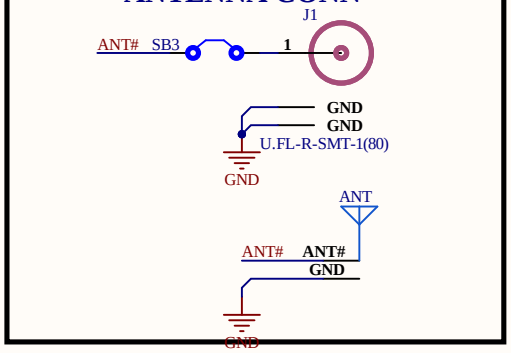


C36 may be disconnected if not needed

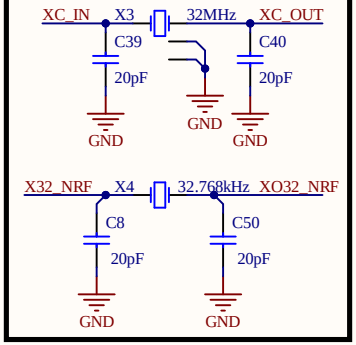
BLE TRANCIEVER POWER BANK



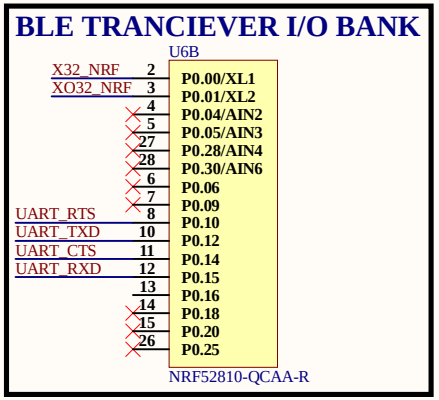
ANTENNA CONN



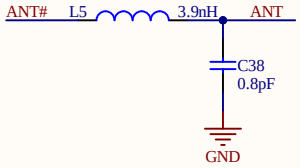
CRYSTALS

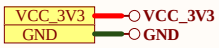


BLE TRANCIEVER I/O BANK



ANTENNA SUPPORTING CIRCUIT

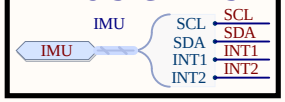




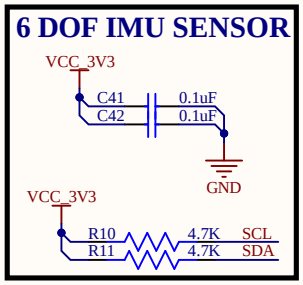
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Title	ArmBand-Control-Board.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	2-10-2025	Sheet 6 of	8
File:	D:\eSpark-Consultants\..\BLE-Section.SchDoc	Drawn By:	Huzaifa Ali

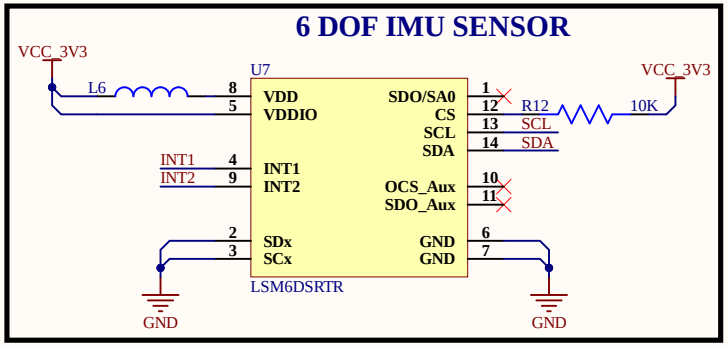
IMU SIGNALS



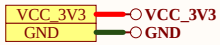
6 DOF IMU SENSOR



6 DOF IMU SENSOR

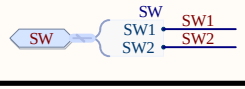


CS logic 1 to enable I2C and disable SPI mode.

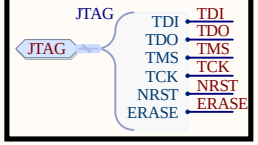


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	Size: A4	Number: 1	Revision: 1
	Date: 2-10-2025	Sheet 7 of 8	
	File: D:\eSpark-Consultants\..\IMU-Section.SchDoc	Drawn By: Huzaifa Ali	

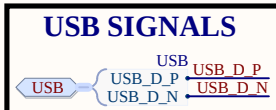
SWITCH SIGNALS



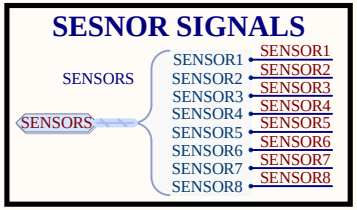
JTAG SIGNALS



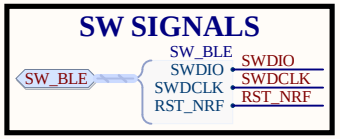
USB SIGNALS



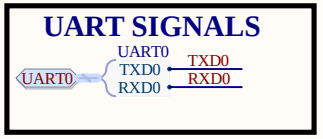
SESNOR SIGNALS

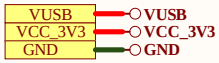


SW SIGNALS

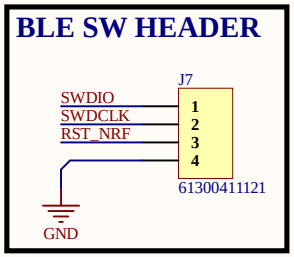


UART SIGNALS

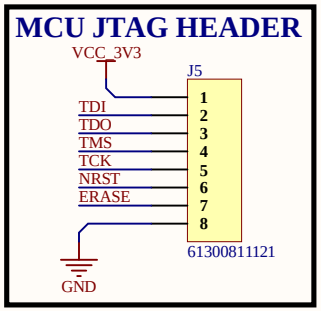




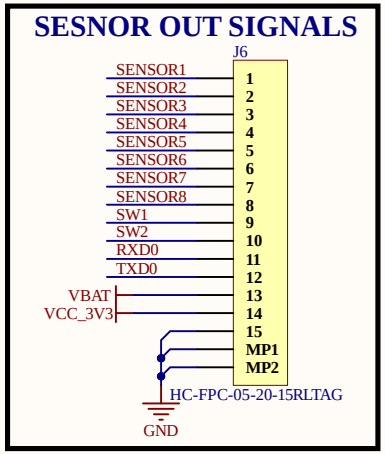
BLE SW HEADER



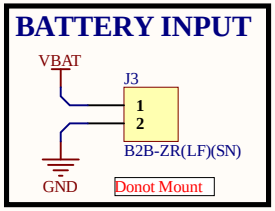
MCU JTAG HEADER



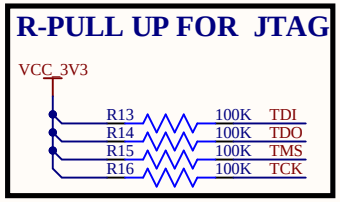
SESNOR OUT SIGNALS



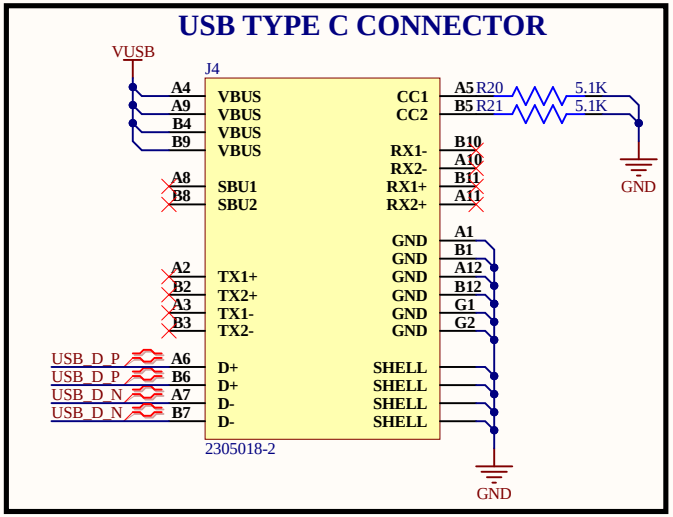
BATTERY INPUT



R-PULL UP FOR JTAG



USB TYPE C CONNECTOR



Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: ArmBand-Control-Board.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 2-10-2025	Sheet 8 of 8	
	File: D:\eSpark-Consultants\..\Connector-Section.DSnDrawn By:		Huzaifa Ali